# Benchmark Comparison with Standard Longitudinal Models

## 1 Benchmark comparison with standard longitudinal models

We compared the default gamlss.longitudinal RS-joint fit against glm and geepack GEE baselines across Gaussian, Gamma, binary, and Poisson longitudinal simulations. The GEE baselines used exchangeable, AR(1), and unstructured working correlation. GAM and GLMM rows were removed from this report because GAM duplicated the fixed-effect glm baseline here, while GLMM targets a conditional mixed-model estimand that is less directly comparable for this marginal response-mean benchmark. The Poisson results are complete for all but the covariate-dependent high-correlation setting, where 51 replicates were available.

Each simulation used four repeated observations per subject and the same fixed-effect marginal mean formula, response $\sim x + z_{null}$. The covariate $x$ was the active slope used for coefficient calibration, while $z_{null}$ was included as a true-null covariate for false-positive-rate checks. The benchmark deliberately excludes low-correlation scenarios and focuses on moderate to high within-subject dependence.

Table 1: Simulation scenarios included in the benchmark. External scenarios were generated independently of gamlss.longitudinal using a Gaussian copula and fixed correlation matrix. Internal scenarios used the package longitudinal simulator to vary the adjacent dependence parameter over time or by covariate value.

| Scenario | Simulator | True dependence | Level | Detail |
|---|---|---|---|---|
| External exchangeable, moderate | external | exchangeable | moderate | Gaussian copula, rho=0.45 |
| External exchangeable, high | external | exchangeable | high | Gaussian copula, rho=0.75 |
| External AR(1), moderate | external | ar1 | moderate | Gaussian copula, rho=0.55 |
| External AR(1), high | external | ar1 | high | Gaussian copula, rho=0.78 |
| Internal time-varying adjacent | internal | time_varying_adjacent | moderate_high | Gaussian copula, adjacent tau=0.35, 0.55, 0.70 |
| Internal covariate-dependent adjacent | internal | covariate_dependent_adjacent | moderate_high | Gaussian copula, adjacent tau=inverse-logit(logit(0.55)+1.1x) |

Table 2: Number of scenario-family cells in which each model had the best value for each benchmark metric. Larger counts indicate broader dominance across the simulation grid.

| Metric | gamlss.longitudinal | geepack exch. | geepack AR(1) | geepack unstr. | glm |
|---|---|---|---|---|---|
| Response mean RMSE | 10 | 0 | 8 | 0 | 6 |
| 90th percentile MAE | 20 | 10 | 7 | 7 | 9 |
| 95% prediction interval coverage | 19 | 20 | 20 | 18 | 20 |
| Upper-tail error | 7 | 3 | 7 | 7 | 0 |
| Runtime | 0 | 0 | 0 | 0 | 24 |

The revised comparison focuses on marginal exponential-family baselines. The glm rows show the cost of ignoring within-subject correlation for inference, while the geepack rows show how much can be recovered with standard working-correlation choices. The unstructured geepack row is included as a flexible working-correlation comparator; 2349 of 2351 unstructured GEE fits completed successfully, with two timeout/failure rows retained as failed benchmark attempts.

Table 3: Median response mean RMSE by response family and model. Lower values are better.

| Family | gamlss.longitudinal | geepack exch. | geepack AR(1) | geepack unstr. | glm |
|---|---|---|---|---|---|
| Gaussian | 0.106 | 0.108 | 0.105 | 0.119 | 0.108 |
| Gamma | 0.093 | 0.095 | 0.094 | 0.102 | 0.095 |
| Binary | 0.045 | 0.046 | 0.046 | 0.052 | 0.046 |
| Poisson | 0.193 | 0.189 | 0.191 | 0.212 | 0.189 |

Table 4: Average coefficient-level inference calibration across the benchmark grid. The SE ratio compares the model-based standard error with the empirical Monte Carlo standard deviation for the slope coefficient; values near 1 are best. CI coverage is the empirical coverage of nominal 95% intervals for the slope coefficient. The false-positive rate is computed for the null coefficient.

| Method | SE ratio | 95% CI coverage | Null false-positive rate |
|---|---|---|---|
| gamlss.longitudinal | 0.967 | 0.933 | 0.063 |
| geepack exchangeable | 1.003 | 0.943 | 0.059 |
| geepack AR(1) | 1.005 | 0.945 | 0.062 |
| geepack unstructured | 0.883 | 0.940 | 0.058 |
| glm | 0.635 | 0.772 | 0.238 |

Table 5: Median elapsed time in seconds by response family and model. Lower values are faster.

| Family | gamlss.longitudinal | geepack exch. | geepack AR(1) | geepack unstr. | glm |
|---|---|---|---|---|---|
| Gaussian | 3.02 | 0.02 | 0.02 | 0.02 | 0.00 |
| Gamma | 4.08 | 0.02 | 0.03 | 0.02 | 0.01 |
| Binary | 16.34 | 0.02 | 0.03 | 0.02 | 0.01 |
| Poisson | 52.73 | 0.01 | 0.02 | 0.02 | 0.00 |

# A Metric-by-metric review tables

The appendix repeats the family-by-model layout for every numeric metric in the saved review table.

Table 6: Median response-mean RMSE by response family and model. Lower values are better.

| Family | gamlss.longitudinal | geepack exch. | geepack AR(1) | geepack unstr. | glm |
|---|---|---|---|---|---|
| Gaussian | 0.106 | 0.108 | 0.105 | 0.119 | 0.108 |
| Gamma | 0.093 | 0.095 | 0.094 | 0.102 | 0.095 |
| Binary | 0.045 | 0.046 | 0.046 | 0.052 | 0.046 |
| Poisson | 0.193 | 0.189 | 0.191 | 0.212 | 0.189 |

Table 7: Median 90th percentile MAE by response family and model. Lower values are better.

| Family | gamlss.longitudinal | geepack exch. | geepack AR(1) | geepack unstr. | glm |
|---|---|---|---|---|---|
| Gaussian | 0.097 | 0.100 | 0.098 | 0.112 | 0.100 |
| Gamma | 0.127 | 0.136 | 0.134 | 0.151 | 0.136 |
| Binary | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| Poisson | 0.183 | 0.200 | 0.205 | 0.225 | 0.200 |

Table 8: Median upper-tail probability error at the true 90th percentile by response family and model. Values closer to zero are better.

| Family | gamlss.longitudinal | geepack exch. | geepack AR(1) | geepack unstr. | glm |
|---|---|---|---|---|---|
| Gaussian | -0.002 | -0.002 | -0.001 | -0.002 | -0.002 |
| Gamma | -0.001 | -0.001 | -0.001 | -0.002 | -0.001 |
| Binary | 0.000 | – | – | – | – |
| Poisson | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 |

Table 9: Median empirical 95% prediction interval coverage by response family and model. Values closer to 0.95 are better.

| Family | gamlss.longitudinal | geepack exch. | geepack AR(1) | geepack unstr. | glm |
|---|---|---|---|---|---|
| Gaussian | 0.950 | 0.950 | 0.950 | 0.950 | 0.950 |
| Gamma | 0.950 | 0.950 | 0.950 | 0.950 | 0.950 |
| Binary | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| Poisson | 0.985 | 0.985 | 0.985 | 0.985 | 0.985 |

Table 10: Median negative log score by response family and model. Lower values are better.

| Family | gamlss.longitudinal | geepack exch. | geepack AR(1) | geepack unstr. | glm |
|---|---|---|---|---|---|
| Gaussian | 1.412 | 1.412 | 1.412 | 1.411 | 1.412 |
| Gamma | 1.232 | 1.232 | 1.232 | 1.232 | 1.232 |
| Binary | 0.638 | 0.638 | 0.639 | 0.639 | 0.638 |
| Poisson | 1.924 | 1.924 | 1.924 | 1.922 | 1.924 |

Table 11: Median elapsed time in seconds by response family and model. Lower values are faster.

| Family | gamlss.longitudinal | geepack exch. | geepack AR(1) | geepack unstr. | glm |
|---|---|---|---|---|---|
| Gaussian | 3.02 | 0.02 | 0.02 | 0.02 | 0.00 |
| Gamma | 4.08 | 0.02 | 0.03 | 0.02 | 0.01 |
| Binary | 16.34 | 0.02 | 0.03 | 0.02 | 0.01 |
| Poisson | 52.73 | 0.01 | 0.02 | 0.02 | 0.00 |

Table 12: Median number of replicates contributing to each response family and model summary.

| Family | gamlss.longitudinal | geepack exch. | geepack AR(1) | geepack unstr. | glm |
|---|---|---|---|---|---|
| Gaussian | 100 | 100 | 100 | 100 | 100 |
| Gamma | 100 | 100 | 100 | 100 | 100 |
| Binary | 100 | 100 | 100 | 100 | 100 |
| Poisson | 100 | 100 | 100 | 100 | 100 |

Table 13: Mean slope standard-error ratio by response family and model. Values closer to one are better.

| Family | gamlss.longitudinal | geepack exch. | geepack AR(1) | geepack unstr. | glm |
|---|---|---|---|---|---|
| Gaussian | 0.956 | 0.986 | 0.987 | 0.994 | 0.605 |
| Gamma | 0.997 | 1.026 | 1.033 | 0.830 | 0.629 |
| Binary | 0.934 | 0.982 | 0.988 | 0.917 | 0.672 |
| Poisson | 0.979 | 1.016 | 1.012 | 0.792 | 0.632 |

Table 14: Mean empirical 95% confidence interval coverage for the slope coefficient by response family and model. Values closer to 0.95 are better.

| Family | gamlss.longitudinal | geepack exch. | geepack AR(1) | geepack unstr. | glm |
|---|---|---|---|---|---|
| Gaussian | 0.930 | 0.935 | 0.932 | 0.952 | 0.757 |
| Gamma | 0.942 | 0.952 | 0.955 | 0.945 | 0.760 |
| Binary | 0.925 | 0.948 | 0.957 | 0.927 | 0.807 |
| Poisson | 0.937 | 0.935 | 0.935 | 0.935 | 0.766 |

Table 15: Mean null-effect standard-error ratio by response family and model. Values closer to one are better.

| Family | gamlss.longitudinal | geepack exch. | geepack AR(1) | geepack unstr. | glm |
|---|---|---|---|---|---|
| Gaussian | 1.001 | 1.028 | 1.033 | 1.013 | 0.629 |
| Gamma | 0.951 | 0.973 | 0.975 | 0.779 | 0.595 |
| Binary | 0.927 | 0.941 | 0.940 | 0.961 | 0.642 |
| Poisson | 0.931 | 0.953 | 0.948 | 0.842 | 0.592 |

Table 16: Mean empirical 95% confidence interval coverage for the null coefficient by response family and model. Values closer to 0.95 are better.

| Family | gamlss.longitudinal | geepack exch. | geepack AR(1) | geepack unstr. | glm |
|---|---|---|---|---|---|
| Gaussian | 0.945 | 0.950 | 0.950 | 0.933 | 0.778 |
| Gamma | 0.933 | 0.937 | 0.935 | 0.938 | 0.758 |
| Binary | 0.938 | 0.943 | 0.938 | 0.943 | 0.780 |
| Poisson | 0.932 | 0.935 | 0.930 | 0.953 | 0.731 |

Table 17: Mean null false-positive rate by response family and model. Lower values are better, with nominal two-sided 5% testing targeting 0.05.

| Family | gamlss.longitudinal | geepack exch. | geepack AR(1) | geepack unstr. | glm |
|---|---|---|---|---|---|
| Gaussian | 0.055 | 0.050 | 0.050 | 0.067 | 0.222 |
| Gamma | 0.067 | 0.063 | 0.065 | 0.062 | 0.242 |
| Binary | 0.062 | 0.057 | 0.062 | 0.057 | 0.220 |
| Poisson | 0.068 | 0.065 | 0.070 | 0.047 | 0.269 |

Table 18: Mean gamlss.longitudinal primary fit success rate attached to each response family and model row.

| Family | gamlss.longitudinal | geepack exch. | geepack AR(1) | geepack unstr. | glm |
|---|---|---|---|---|---|
| Gaussian | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| Gamma | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| Binary | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| Poisson | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |

Table 19: Mean gamlss.longitudinal primary fit convergence rate attached to each response family and model row.

| Family | gamlss.longitudinal | geepack exch. | geepack AR(1) | geepack unstr. | glm |
|---|---|---|---|---|---|
| Gaussian | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| Gamma | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| Binary | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| Poisson | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |

Table 20: Median gamlss.longitudinal primary fit elapsed time in seconds attached to each response family and model row.

| Family | gamlss.longitudinal | geepack exch. | geepack AR(1) | geepack unstr. | glm |
|---|---|---|---|---|---|
| Gaussian | 3.02 | 3.02 | 3.02 | 3.02 | 3.02 |
| Gamma | 4.08 | 4.08 | 4.08 | 4.08 | 4.08 |
| Binary | 16.34 | 16.34 | 16.34 | 16.34 | 16.34 |
| Poisson | 52.73 | 52.73 | 52.73 | 52.73 | 52.73 |